Project 3: Discrete Fourier Transform

Results, Plots, and PYNQ-Z2 Demo | Questions 1-9

Recorded results and discussion for the DFT32 and DFT1024 designs, the best DFT1024 design (Q8), its AXI-Stream version (Q9), and the hardware demo on the PYNQ-Z2. Latency times that were not recorded are calculated from cycles.

Timing convention: 100 MHz (10 ns target clock). Latency time = latency cycles x 10 ns. Throughput = 100,000,000 / recorded interval, in transforms per second. Worst-case intervals are used for ranges. Existing time values retain their recorded rounding.

Q1-Q3 | Baseline, table lookup, and interface change

Hardware resources

Design	BRAM_18K	DSP	FF	LUT
Q1: cos/sin	18	195	13,323	16,168
Q2: 2D table	68	640	59,556	92,050
Q3: four arrays	64	640	59,189	91,521

Timing and throughput

Design	Latency (cycles)	Latency time	Interval (cycles)	Throughput (DFT32/s)
Q1: cos/sin	193,095 to 205,383	1.93095-2.05383 ms	193,096 to 205,384	487
Q2: 2D table	291	2.91 us	292	342,466
Q3: four arrays	221	2.21 us	222	450,450

Numerical accuracy

Design	RMSE_R	RMSE_I
Q1: cos/sin	2.99e-6	2.46e-5
Q2: 2D table	2.20e-6	1.83e-6
Q3: four arrays	2.20e-6	1.83e-6

Q1(a) | Recorded discussion of CORDIC accuracy

Changing CORDIC accuracy changes both resource usage and performance. More iterations and wider datapaths increase LUT/FF usage (linearly, if unrolled) and increase per-call latency (linearly, if iterative), while too little accuracy increases RMSE against the golden output and may fail the 0.1 threshold; beyond the accuracy needed to keep RMSE well below threshold, extra iterations only waste area.

Q3(a), (c) | Recorded discussion of the interface change

Splitting the interface into separate input and output arrays removes the temporary arrays and the copy-back loop: the inputs are now read-only and the outputs are accumulated directly, so BRAM falls from 68 to 64 and latency falls from 291 to 221 cycles. However, DSP (640) and LUT (91521) are essentially unchanged and still exceed the PYNQ-Z2 capacity, because Vitis HLS automatically pipelines the outer loop, completely unrolls both inner loops (factor 32), and completely partitions the coefficient tables, instantiating 128 floating-point multipliers. The interface change is therefore a structural improvement, not a resource optimization; controlling the degree of parallelism is left to the array partitioning in Q4.

Q4 | DFT32 array partitioning

Block partition factors: 1, 2, 4, 8, 16, and 32. The supplied plots identify a pipeline target of II = 1; achieved values are listed below.

Q4(a) | Recorded results

Block factor	Achieved II	BRAM_18K	DSP	FF	LUT	Latency (cycles)	Latency time	Interval (cycles)	Throughput (DFT32/s)
1	16	0	559	214,186	174,643	187	1.87 us	16	6,250,000
2	8	0	1,136	324,466	293,723	178	1.78 us	8	12,500,000
4	4	0	2,265	434,545	498,761	174	1.74 us	4	25,000,000
8	2	0	4,522	644,577	903,786	172	1.72 us	2	50,000,000
16	1	0	8,908	954,603	1,564,610	171	1.71 us	1	100,000,000
32	1	0	8,908	952,205	1,564,546	168	1.68 us	1	100,000,000

All six rows: RMSE_R = 2.20e-6; RMSE_I = 1.83e-6.

Q4(b) | Resource utilization

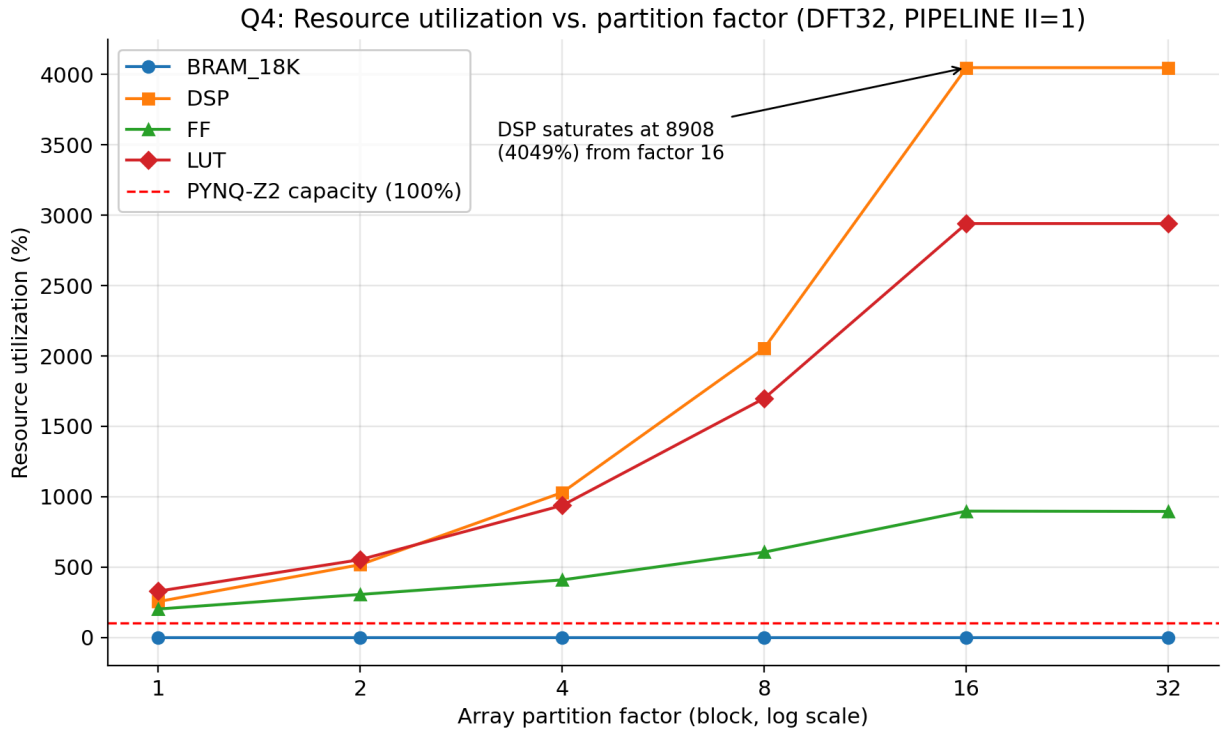


Figure 1. Original resource-utilization plot versus block partition factor.
Source: q4_resource_utilization_vs_partition.png

Q4 | Latency and throughput plots

Q4(c) | Supplied plots

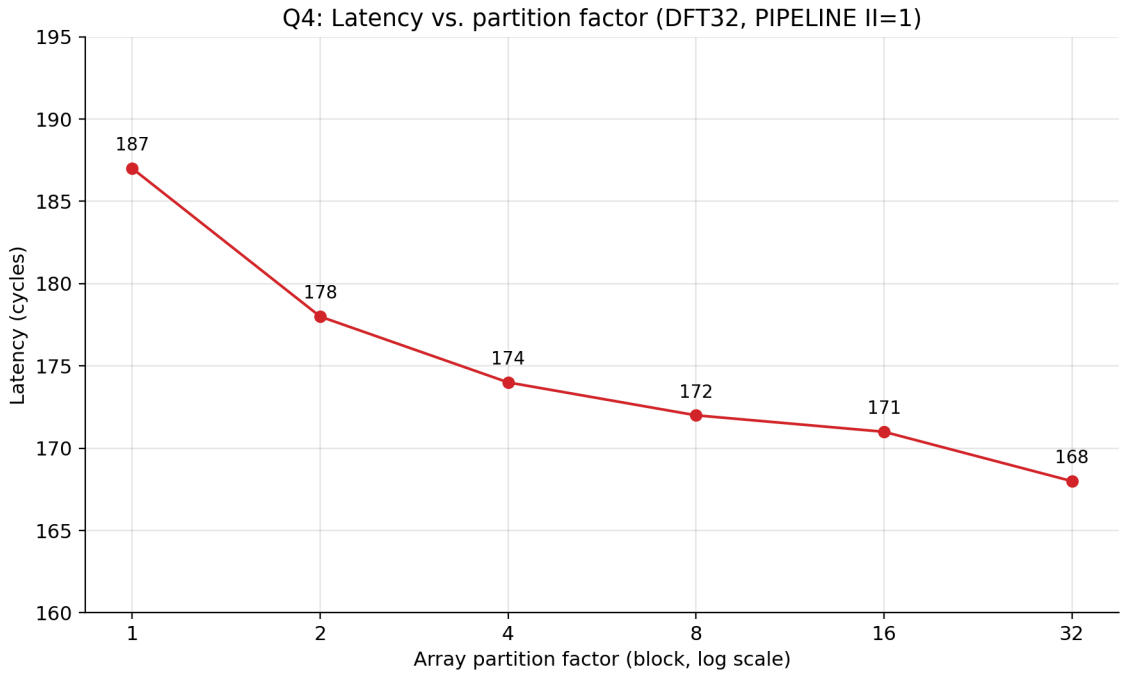


Figure 2. Latency versus block partition factor.
Source: q4_latency_vs_partition.png

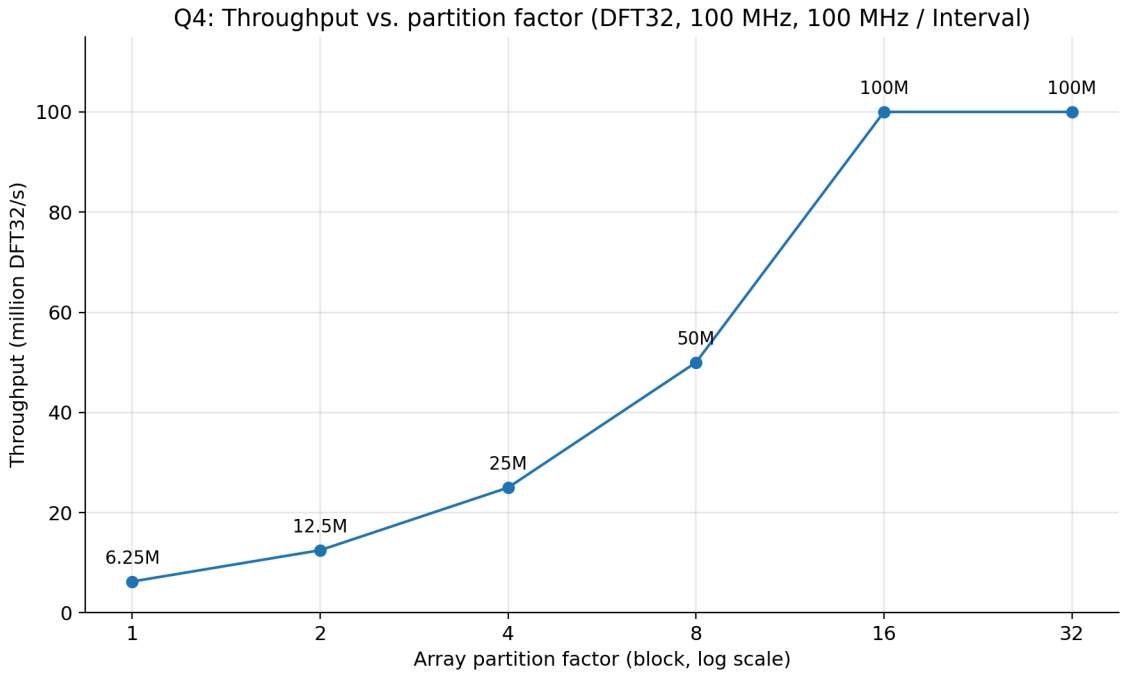


Figure 3. Throughput versus block partition factor, using 100 MHz / interval.
Source: q4_throughput_vs_partition.png

Q5 | DFT32 inner-loop unrolling

Recorded setup: block partition factor 16, with inner-loop unrolling. The supplied plots label these experiments as having no PIPELINE pragma.

Q5(a) | Recorded results

Unroll factor	Achieved II	BRAM_18K	DSP	FF	LUT	Latency (cycles)	Latency time	Interval (cycles)	Throughput (DFT32/s)
1	207	64	640	57,515	91,252	206	2.06 us	207	483,092
2	191	64	1,280	112,432	182,312	190	1.90 us	191	523,560
4	183	0	2,560	229,933	365,224	182	1.82 us	183	546,448
8	179	0	5,120	457,259	729,256	178	1.78 us	179	558,660
16	176	0	6,104	590,477	988,911	175	1.75 us	176	568,182
32	13,249	0	5	51,005	79,316	13,248	132.48 us	13,249	7,548

All six rows: RMSE_R = 2.20e-6; RMSE_I = 1.83e-6.

Recorded observation: "32 unroll and block factor=16 is contradicting." The factor-32 values and the plot annotation are preserved as supplied.

Q5(b) | Resource utilization

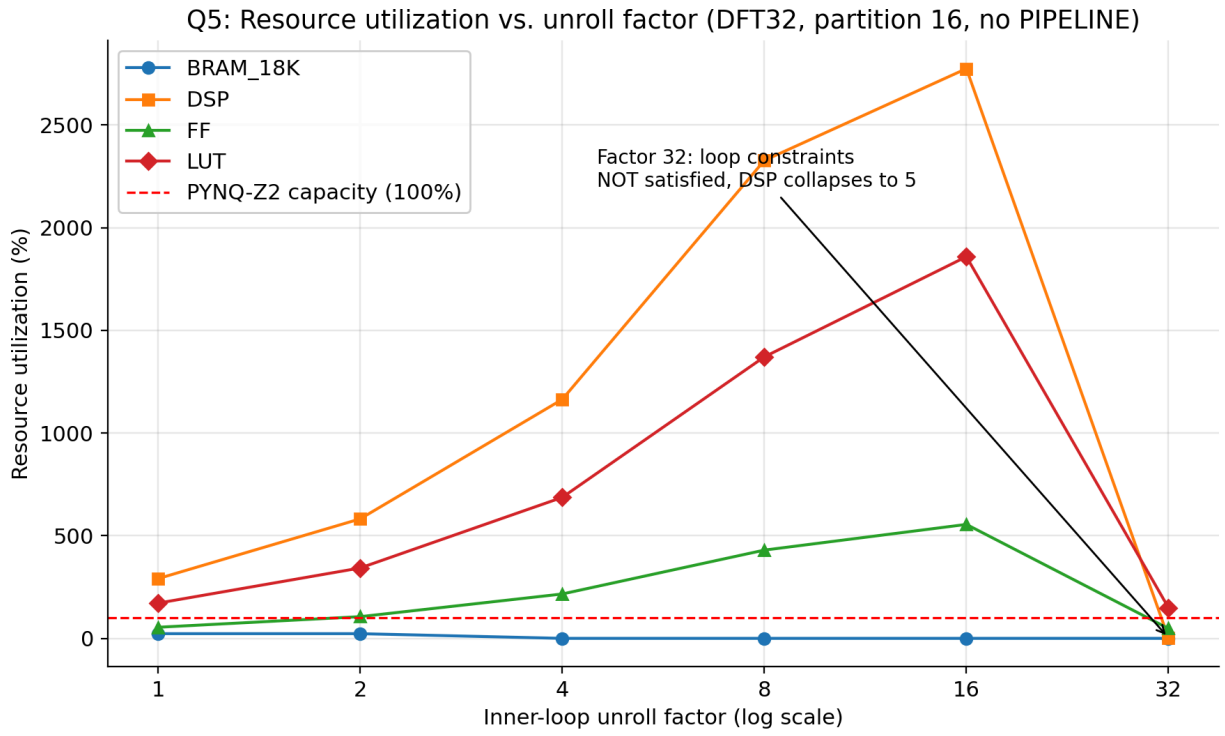


Figure 4. Original resource-utilization plot versus inner-loop unroll factor.
Source: q5_resource_utilization_vs_unroll.png

Q5 | Latency and throughput plots

Q5(c) | Supplied plots

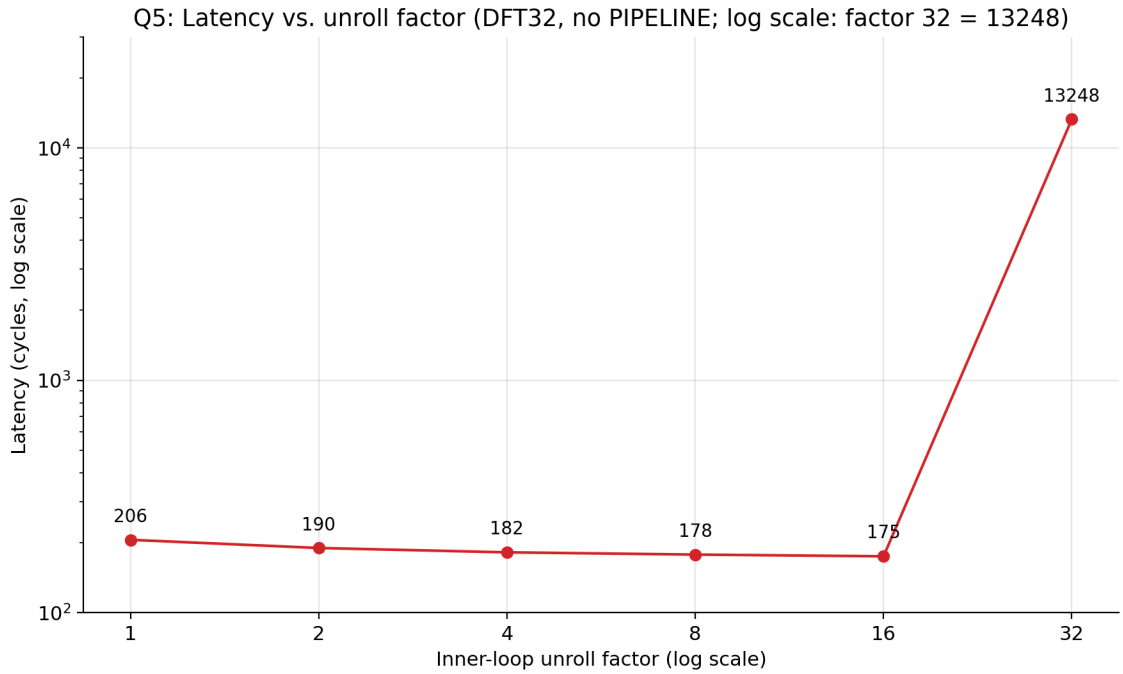


Figure 5. Latency versus inner-loop unroll factor; the latency axis uses a logarithmic scale.
Source: q5_latency_vs_unroll.png

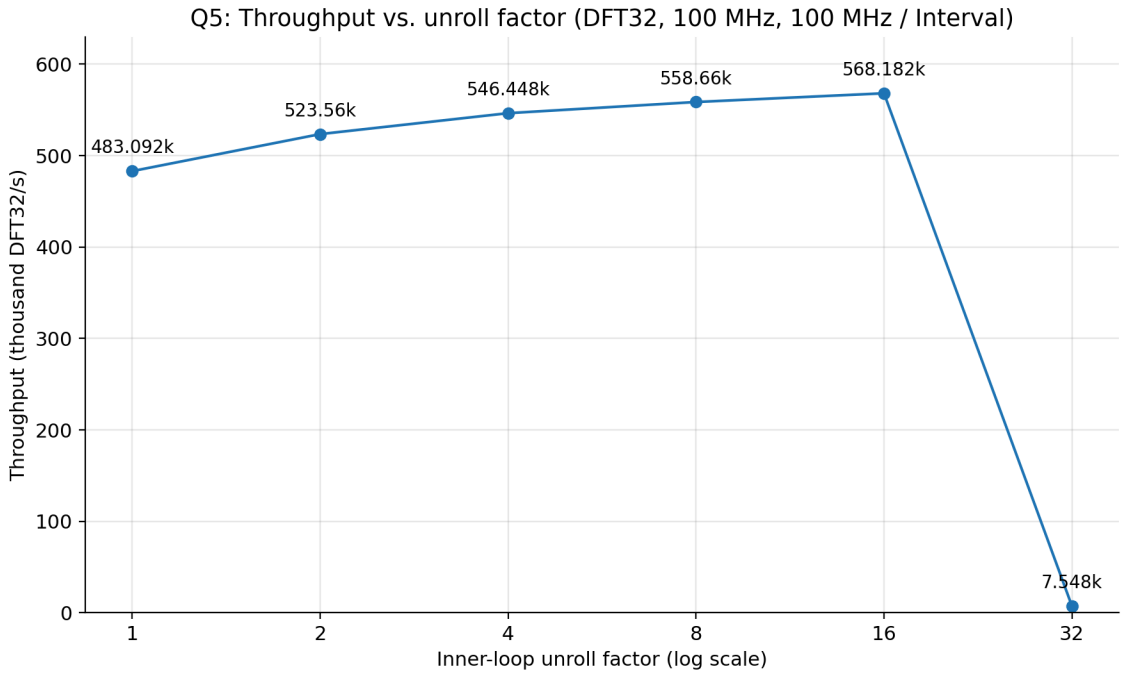


Figure 6. Throughput versus inner-loop unroll factor, using 100 MHz / interval.
Source: q5_throughput_vs_unroll.png

Q6-Q7 | DFT1024 recorded results

Q6(a): baseline using cos/sin. Q6(b): precomputed table. Q7: reverse the loop order.

Hardware resources

Design	BRAM_18K	DSP	FF	LUT
Q6(a): cos/sin	16	195	13,325	15,861
Q6(b): 1D table	4	5	968	1,413
Q7: loop order	4	20	2,183	3,380

Timing and throughput

Design	Latency (cycles)	Latency time	Interval (cycles)	Throughput (DFT1024/s)
Q6(a): cos/sin	211,827,713 to 224,410,625	2.118-2.244 s	211,827,714 to 224,410,626	0.446
Q6(b): 1D table	12,614,657	0.126 s	12,614,658	7.927
Q7: loop order	1,049,624	10.496 ms	1,049,625	95.27

Numerical accuracy

Design	RMSE_R	RMSE_I
Q6(a): cos/sin	2.17e-3	5.60e-3
Q6(b): 1D table	1.85e-3	1.31e-3
Q7: loop order	1.85e-3	1.31e-3

Q6(a) throughput uses the recorded worst-case interval: 100,000,000 / 224,410,626 = approximately 0.446 DFT1024/s.

Q8 | Best DFT1024 design

Starting from the loop-interchanged Q7 design, the sweep below varies U, the number of output bins processed per cycle. All four runs use the same target (xc7z020clg400-1, 10 ns clock). The main loop reaches II = 1 in every run, so its length is 1024 x (1024 / U) + 1 cycles. The estimated clock period is 7.256 ns with 2.70 ns uncertainty in all four runs.

Q8(a) | Sweep results

U	II	BRAM_18K	DSP	FF	LUT	Latency (cycles)	Latency time	Interval (cycles)	Throughput (DFT1024/s)
2	1	16 (5%)	40 (18%)	4,149 (3%)	6,706 (12%)	526,364	5.264 ms	526,365	189.98
4	1	28 (10%)	80 (36%)	8,134 (7%)	12,913 (24%)	264,222	2.642 ms	264,223	378.47
8 (best)	**1**	**52 (18%)**	**160 (72%)**	**15,970 (15%)**	**25,496 (47%)**	**133,151**	**1.332 ms**	**133,152**	**751.02**
16 (does not fit)	1	100 (35%)	320 (145%)	31,397 (29%)	50,532 (94%)	67,617	0.676 ms	67,618	1,478.90

Accuracy for the selected design (U = 8) is unchanged from Q7: RMSE_R = 1.85e-3 and RMSE_I = 1.31e-3, because each output bin is still accumulated over the same inputs in the same order. Throughput = 100,000,000 / interval (cycles).

Q8(b) | Resource utilization and throughput plots

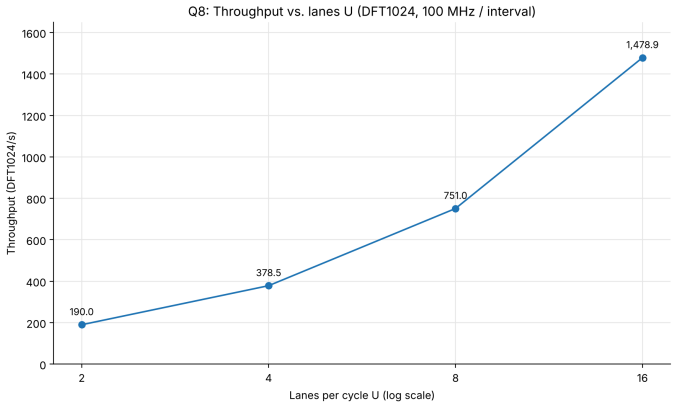


Figure 7. Throughput versus lanes U, using 100 MHz / interval.
Source: q8_throughput_vs_U.png

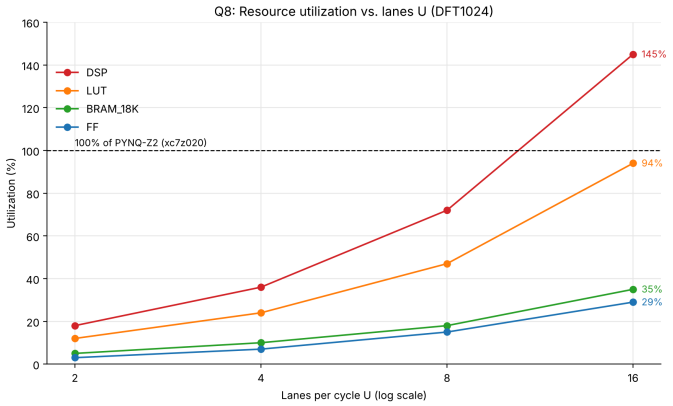


Figure 8. Resource utilization versus lanes U. U = 16 exceeds the DSP capacity of the PYNQ-Z2.
Source: q8_utilization_vs_U.png

Q8 | Optimization method and discussion

Q8(c) | How the design was built

The inner j loop was unrolled by U and the real and imaginary accumulators were partitioned cyclically by U, so each lane owns its own accumulator bank. The twiddle index (i x j) mod 1024 changes bank with i, so array partitioning cannot make the table reads conflict-free. Instead, every pair of lanes gets its own copy of the cosine and sine tables, one copy per two read ports (U / 2 copies of each table). Accumulation happens in on-chip buffers that are copied to the output arrays at the end. The loop order from Q7 (i outer, j inner) removes the floating-point accumulate dependency, so the pipelined main loop reaches II = 1 for every U. No FFT symmetry was used; the twiddle index is computed as (i x j) & 1023.

Q8(d) | Scaling and resource cost

Throughput scales almost linearly with U. The speedup over the Q7 interval of 1,049,625 cycles is 1.99x, 3.97x, 7.88x, and 15.5x for U = 2, 4, 8, and 16. The shortfall from ideal comes from the initialization and copy-out loops, which cost about 2,050 cycles regardless of U; that share grows from 0.4% of the interval at U = 2 to 3% at U = 16.

Each lane costs exactly 20 DSPs (four floating-point multipliers at 3 DSPs each, plus four add/subtract units at 2 DSPs each) and roughly 3,000 LUTs. The PYNQ-Z2 has 220 DSPs, so at most 11 lanes fit by DSP count alone. U must divide 1024 evenly, which restricts the choices to powers of two; U = 16 needs 320 DSPs (145%) and 94% of the LUTs and cannot be implemented. **U = 8 is therefore the best design:** 133,152 cycles per transform, 751.02 DFT1024/s, using 72% of the DSPs and 47% of the LUTs.

Q8(e) | How throughput was calculated

For every synthesis result, throughput = clock frequency / initiation interval of the whole function = 100,000,000 Hz / interval (cycles), with the 10 ns target clock. The interval reported by Vitis HLS (latency + 1, since the function does not overlap successive calls) is used, not the latency. The same convention is used for all sections of this report.

Q9 | AXI-Stream interface

The Q8 design (U = 8) was converted to the streaming interface: the arrays became four **hls::stream<hls::axis<float,0,0,0>>** ports (real input, imaginary input, real output, imaginary output), each carrying 32-bit data with TKEEP, TSTRB, and TLAST. The function control uses an AXI-Lite slave (**s_axilite** on the return port). Each input stream is read once into a local buffer, the Q8 compute core runs on the buffers, and the results are written to the output streams with TLAST set on the final packet.

Q9(a) | Q8 versus Q9

Design	BRAM_18K	DSP	FF	LUT	Latency (cycles)	Interval (cycles)	Throughput (DFT1024/s)
Q8: arrays, U = 8	52	160	15,970	25,496	133,151	133,152	751.02
Q9: streams, U = 8	56	160	15,944	25,775	133,152	133,153	751.02

Q9 utilization: BRAM 20%, DSP 72%, FF 14%, LUT 48%. Estimated clock period 7.256 ns (10 ns target, 2.70 ns uncertainty).

Q9(b) | Loop timing

Loop	Role	Interval (cycles)
init_read	Read one packet from each input stream; zero accumulators; fill twiddle table copies	1,025
outer / inner	Compute: 1024 x (1024 / 8) iterations at II = 1	131,073
write_out	Write results to the output streams; TLAST on the last packet	1,025

Q9(c) | Interfaces in the synthesis report

Port group	Protocol	Signals
real_sample, imag_sample	AXI-Stream, in	TDATA[31:0], TVALID, TREADY, TLAST, TKEEP[3:0], TSTRB[3:0]
real_op, imag_op	AXI-Stream, out	TDATA[31:0], TVALID, TREADY, TLAST, TKEEP[3:0], TSTRB[3:0]
s_axi_control	AXI-Lite slave	ap_start / ap_done / ap_idle control register, interrupt

Q9(d) | Discussion

The streaming version has the same compute performance as Q8. Throughput is unchanged at 751.02 DFT1024/s because the interval grows by only one cycle at the Q8 U = 8 setting (133,152 to 133,153). BRAM rises from 52 to 56 because the two 1,024-word input buffers hold the samples while the compute loop re-reads each sample 128 times. The input and output phases (about 2,050 cycles, or 1.5% of the interval) are not overlapped with the compute loop, so a DATAFLOW arrangement could hide them at the cost of extra buffering. That was not needed here because the compute loop dominates.

The C simulation passes with the streaming testbench: RMSE_R = 1.85e-3, RMSE_I = 1.31e-3 (same as Q7 and Q8), no TLAST errors, and no leftover packets. Vitis HLS reports a maximum stream depth of 1,024 during C simulation, which is expected because the testbench writes all 1,024 inputs before calling the function; the hardware streams use AXI-Stream back-pressure.

Demo | PYNQ-Z2 with two AXI DMAs

The Q9 IP was exported from Vitis HLS and integrated in Vivado 2024.2 for the PYNQ-Z2 (xc7z020clg400-1).

Demo(a) | Block design

Block	Configuration
ZYNQ7 Processing System	FCLK_CLK0 = 100 MHz; M_AXI_GP0 for control; S_AXI_HP0 enabled for DMA memory access
axi_dma_0	Simple register mode (scatter-gather off), read and write channels enabled; carries the real input and real output streams
axi_dma_1	Same configuration; carries the imaginary input and imaginary output streams
dft_0	Q9 IP; AXI-Lite control from GP0; four AXI-Stream ports connected to the two DMAs
smartconnect_mem	One SmartConnect with four slave ports (both memory-mapped channels of both DMAs) feeding S_AXI_HP0

Implementation with the 100 MHz clock met timing: worst setup slack +0.691 ns (WNS), worst hold slack +0.011 ns (WHS), 0 failed nets, 0 critical warnings.

Demo(b) | Operation

The notebook loads the overlay (dft.bit and dft.hwh) and allocates four contiguous 1,024-word float32 buffers. Each transform sets AP_START in the IP, starts both receive DMA transfers first, then starts both send transfers, and waits for all four to finish. Receive channels are started before the send channels so that the output stream always has a destination.

Demo(c) | Measured results

Quantity	Value
Input	real = 0, 1, ..., 1023; imaginary = 0
RMSE against NumPy fft (real / imaginary)	1.93e-3 / 8.59e-3
Average time per DFT1024 (100 runs)	3.561 ms
Measured system throughput	281 DFT1024/s
Accelerator estimate from synthesis	1.332 ms; 751.02 DFT1024/s

Measured throughput = 1 / average time per transform, including the DMA transfers and the Python driver code. It is lower than the synthesis estimate of 751.02 DFT1024/s because the synthesis estimate covers only the accelerator (1.332 ms). The remaining time, about 2.2 ms, is the cost of setting up and waiting for four DMA transfers from Python. The RMSE is measured against NumPy's double-precision FFT and is consistent with a 32-bit float hardware implementation; it is well below the 0.1 threshold.

Demo video: https://drive.google.com/file/d/1ag_OzxPy7PhNTbTmD_pnfe8eENj-vr2Y/view?usp=drive_link

Sources and scope

Measurement source: Vitis HLS 2024.2 synthesis reports (dft_csynth.rpt) for each design and the Vivado 2024.2 implementation log for the demo overlay. Figure sources: the PNG filenames listed beneath Figures 1-8.

Project requirements: pp4fpgas Project 3: Discrete Fourier Transform. Reference consulted: Parallel Programming for FPGAs, DFT chapter.

GitHub repository: https://github.com/UCSD-237C-2026/wes-237c-project-3-dft-group-10